MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, TIMING CONTROL UNIT, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic silicon, CMOS, timing control unit (TCU) microcircuit. One product assurance class is provided and is reflected in the complete Part or Identifying Number (PIN) (see 6.7).

1.2 Classification.

1.2.1 Device type. The device type shall be as follows:

Device type	Frequency	Circuit
01	10.0 MHz	Timing control unit

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1.290" x .610" x .225"), dual-in-line package

1.3 Absolute maximum ratings.

Supply voltage range	-0.3 V dc to +7.0 V dc
Input voltage range	-0.5 V to V_{CC} +0.5 V
Output voltage range	-0.5 V to V_{CC} +0.5 V
Maximum power dissipation	1.0 W
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC})	See MIL-M-38510, appendix C
Maximum junction temperature (T_J)	130°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.75 V dc to 5.25 V dc
Case operating temperature range	-55°C to +125°C
Minimum high level input voltage (V_{IH}) except XIN and $\overline{RSTI}$	2.4 V
Maximum low level input voltage (V_{IL}) except XIN and $\overline{RSTI}$	0.8 V
Minimum high level input voltage XIN	3.8 V
Maximum low level input voltage XIN	1.0 V
Oscillation frequency	10.0 MHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$, V_{CC} = 5.0 V ±5%	Limits Min	Limits Max	Unit
High level output voltage All outputs except XOUT, PHI1, PHI2	V_{OH}			V_{CC} = 4.75 V, I_{OH} = -1 mA 2/	4.27		V
				V_{CC} = 5.25 V, I_{OH} = -1 mA 2/	4.72		
Low level output voltage All outputs except XOUT, PHI1, PHI2	V_{OL}			I_{OL} = 2 mA 2/		0.5	V
High level input voltage all except $\overline{RSTI}$ and XIN	V_{IH}			2/	2.4	5.75	V
Low level input voltage all except $\overline{RSTI}$ and XIN	V_{IL}			2/	-0.5	0.8	V
$\overline{RSTI}$ rising threshold voltage	V_{T+}			2/	2.0	3.5	V
$\overline{RSTI}$ hysteresis voltage	V_{HYS}			2/	0.8	1.8	V
XIN input high voltage	V_{XH}			V_{CC} = 4.75 V 2/	3.8		V
				V_{CC} = 5.25 V 2/	4.2		
XIN input low voltage	V_{XL}			V_{CC} = 4.75 V 2/		0.94	V
				V_{CC} = 5.25 V 2/		1.05	
Low level output voltage PHI1 and PHI2	V_{OL}			I_{OL} = 1 mA 2/		0.5	V
High level output voltage PHI1 and PHI2	V_{OH}			V_{CC} = 4.75 V 2/	4.51		V
				V_{CC} = 5.25 V 2/	4.98		V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$ $V_{CC} = 5.0$ V ±8%	Limits Min	Limits Max	Unit
Input low current	I_{IL}			$V_{IN} = 0.4$ V 2/	-20		µA
Input high current	I_{IH}			$V_{IN} = 5.25$ V 2/		20	µA
Three-state output leakage current high ($\overline{RD}$, $\overline{WR}$)	I_{OZH}			$V_{OUT} = 5.25$ V 2/		20	µA
Three-state output leakage current low ($\overline{RD}$, $\overline{WR}$)	I_{OZL}			$V_{OUT} = 0.4$ V 2/	-20		µA
Supply current	I_{CC}			f = 10 MHz 2/		120	mA
Clock period	t_{CP} 3/	A	9	PHI1 R.E. to next PHI1 R.E. 2/	100		ns
Clock high time for PHI1 and PHI2	t_{CLh} 3/	A	10	90% V_{CC} on PHI1 R.E. to 90% V_{CC} on PHI1 F.E. and 90% V_{CC} on PHI2 R.E. to 90% V_{CC} on PHI2 F.E. 2/	35	48	ns
Clock low time	t_{CL} 3/	A	13	10% V_{CC} on PHI1 F.E. to 10% V_{CC} on PHI1 R.E. 2/	43	60	ns
Clock pulse width PHI1	$t_{CLw(1)}$ 3/	A	53	At 2.0 V on PHI1 2/	40	52	ns
Clock pulse width PHI2	$t_{CLw(2)}$ 3/	A	54	At 2.0 V on PHI2 2/	40	52	ns
PHI1, PHI2 asymmetry	t_{CLwas}			At 2.0 V on PHI1, PHI2 2/	-5	5	ns
Clock rise time	t_{CLR} 3/	A	12	10% V_{CC} on PHI1 R.E. to 90% V_{CC} on PHI1 R.E.		8 4/	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$ $V_{CC} = 5.0\ V \pm 5\%$	Limits Min	Limits Max	Unit
Clock fall time	t_{CLE} 3/	A	11	90% V_{CC} on PHI1 F.E. to 10% V_{CC} on PHI1 F.E.		8 4/	ns
Clock non-overlap time	t_{nOVl} 3/ 4/	A	14	10% V_{CC} on PHI1 to 10% V_{CC} on PHI2	-2	2	ns
Non-overlap asymmetry	t_{nOVLas} 3/ 4/			At 10% V_{CC} of PHI1, PHI2 2/	-4	4	ns
XIN high time (external output)	t_{Xh}	A	1	At 80% V_{CC} on XIN (both edges) 2/	16		ns
XIN low time (external output)	t_{Xl}	A	2	At 20% V_{CC} on XIN (both edges) 2/	16		ns
XIN to FCLK R.E. delay	t_{XFr}	A	3	80% V_{CC} on XIN R.E. to FCLK R.E. 2/	3	29	ns
XIN to FCLK F.E. delay	t_{XFf}	A	4	20% V_{CC} on XIN F.E. to FCLK F.E. 2/	3	29	ns
XIN to CTTL R.E. delay	t_{XCr} 5/	A	5	80% V_{CC} on XIN R.E. to CTTL R.E. 2/	3	34	ns
XIN to PHI1 R.E. delay	t_{XPr} 3/	A	8	80% V_{CC} on XIN R.E. to PHI1 R.E. 2/	3	32	ns
FCLK to CTTL R.E. delay	t_{FCr} 5/	A	6	FCLK R.E. to CTTL R.E. 2/	-2	6	ns
FCLK to CTTL F.E. delay	t_{FCf}	A	7	FCLK R.E. to CTTL R.E. 2/	-3	4	ns
FCLK to PHI1 R.E. delay	t_{FPr} 3/	B	16	FCLK R.E. to PHI1 R.E. 2/	-1	4	ns
FCLK to PHI1 F.E. delay	t_{FPf} 3/	B	17	FCLK R.E. to PHI1 F.E. 2/	-4	-3	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$, V_{CC} = 5.0 V ±5%	Limits Min	Limits Max	Unit
FCLK to pulse width with crystal	t_{Fw}	B	15	At 50% V_{CC} on FCLK (both edges) 2/	16	25	ns
PHI2 R.E. to CTTL F.E. delay	t_{PCf} 5/	B	18	PHI2 R.E. to CTTL F.E. 2/	-3	4	ns
PHI1 to CTTL R.E. delay	t_{PCr}	B	19	PHI1 R.E. to CTTL R.E. 2/	-3	5	ns
CTTL rise time	t_{CTR}	B	21	10% to 90% V_{CC} on CTTL R.E. 2/		5 4/	ns
CTTL fall time	t_{CTF}	B	22	90% to 10% V_{CC} on CTTL F.E. 2/		5 4/	ns
$\overline{RSTO}$ R.E. delay	t_{RSTr}	F	31	After PHI1 R.E. 2/		21	ns
$\overline{RSTI}$ setup time	t_{RSTs}	F	30	Before PHI1 R.E. 2/	20		ns
$\overline{ADS}$ setup time	t_{ADs}	F	32	Before PHI1 R.E. 2/	30		ns
$\overline{ADS}$ pulse width	t_{ADw}	F	33	$\overline{ADS}$ L.E. + $\overline{ADS}$ T.E. 2/	25		ns
$\overline{DDIN}$ setup time	t_{DDs}	F	34	Before PHI1 R.E. 2/	15		ns
$\overline{TSO}$ L.E. delay	t_{Tf}	C&E	23	After PHI1 R.E. 2/		12	ns
$\overline{TSO}$ T.E. delay	t_{Tr}	C&E	24	After PHI1 R.E. 2/		18	ns
CTTL pulse width	t_{CTw}	A	20	At 50% V_{CC} on CTTL (both edges) 2/	43	51	ns
$\overline{RD}/\overline{WR}$ L.E. delay (fast cycle)	$t_{RWf(F)}$	E	25	After PHI1 R.E. 2/		35	ns
$\overline{RD}/\overline{WR}$ L.E. delay (peripheral cycle)	$t_{RWf(S)}$	C	35	After PHI1 R.E. 2/		25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$, $V_{CC} = 5.0$ V ±5%	Limits Min	Limits Max	Unit
$\overline{RD}/\overline{WR}$ T.E. delay	t_{RWr}	C & E	26	After PHI1 R.E. 2/		25	ns
$\overline{DBE}$ L.E. delay (write cycle)	$t_{DBf(W)}$	C & E	27	After PHI1 R.E. 2/		28	ns
$\overline{DBE}$ L.E. delay (read cycle)	$t_{DBf(R)}$	C & E	28	After PHI2 R.E. 2/		21	ns
$\overline{DBE}$ T.E. delay	t_{DBr}	C & E	29	After PHI2 R.E. 2/		23	ns
$\overline{RD}$, $\overline{WR}$ low level to 3-state	t_{PLZ}	D	36	After $\overline{RWEN}$/SYNC R.E. 2/		30	ns
$\overline{RD}$, $\overline{WR}$ high level to 3-state	t_{PHZ}	D	39	After $\overline{RWEN}$/SYNC R.E. 2/		30	ns
$\overline{RD}$, $\overline{WR}$ 3-state to low level	t_{PZL}	D	37	After $\overline{RWEN}$/SYNC F.E. 2/		25	ns
$\overline{RD}$, $\overline{WR}$ 3-state to high level	t_{PZH}	D	38	After $\overline{RWEN}$/SYNC F.E. 2/		25	ns
$\overline{CWAIT}$ setup time (cycle hold)	$t_{CWs(H)}$	G	41	Before PHI1 R.E. 2/	30		ns
$\overline{CWAIT}$ hold time (cycle hold)	$t_{CWh(H)}$	G	42	After PHI1 R.E. 2/	0		ns
$\overline{CWAIT}$ setup time (wait states)	$t_{CWs(W)}$	G & H	40	Before PHI2 R.E. 2/	10		ns
$\overline{CWAIT}$ hold time (wait states)	$t_{CWh(W)}$	H	44	After PHI2 R.E. 2/	20		ns
$\overline{WAIT}n$ setup time	t_{Ws}	H	45	Before PHI2 R.E. 2/	7		ns
$\overline{WAIT}n$ hold time	t_{Wh}	H	46	After PHI2 R.E. 2/	20		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Fig. 5	Ref. no. 1/	Conditions $-55°C \le T_C \le +125°C$, V_{CC} = 5.0 V ±5%	Limits Min	Limits Max	Unit
PER setup time	t_{Ps}	I	49	Before PHI1 R.E. 2/	7		ns
PER hold time	t_{Ph}	I	48	After PHI1 R.E. 2/	30		ns
RDY delay	t_{Rd}	H & I	47	After PHI2 R.E. 2/		25	ns
SYNC setup time	t_{Sys}	J	50	Before XIN R.E. 2/	6		ns
SYNC hold time	t_{Syh}	J	51	After XIN R.E. 2/	0		ns
CTTL/SYNC inversion delay	t_{CS}	J	52	CTTL (master) to RWEN/SYNC (slave) 2/		10	ns
Capacitance input	C_I			6/		30	pF
Capacitance output	C_O			6/		40	pF
Capacitance clock	C_{clk}			6/		30	pF

1/ The reference number refers to the position where the parameter appears on figure 5.

2/ Timing measurements are made at 10% and 90% of V_{CC}, unless otherwise specified. DC voltages are verified at a test frequency of 1 MHz. AC test voltage levels are V_{IL} = 0.4 V and V_{IH} = 3.0 V except X_{IN} and RSTI which are V_{IL} = 0.5 V and V_{IH} = 4.2 V. All output voltage levels are V_{OL} = 1.6 V and V_{OH} = 2.0 V. AC test are performed at 10 MHz and C_L = 30 pF. R.E. = Rising edge, T.E. = trailing edge, F.E. = falling edge, and L.E. = leading edge.

3/ PHI1 and PHI2 are interchangeable for the following parameters: t_{Cp}, t_{CLh}, t_{CL1}, t_{CLw}, t_{CLR}, t_{CLF}, t_{DVL}, t_{XPr}, t_{FPr}, t_{FPf}.

4/ This limit guaranteed to limit specified herein by characterization data.

5/ Parameters t_{XCr}, t_{FCr}, t_{PCf}, and t_{CTh} are measured with 30 pF load on CTTL.

6/ The capacitance measurements shall be made between the indicated terminal and ground at a frequency of 1 MHz, and T_C = +25°C. The dc bias of the measuring instrument shall be less than ±0.1 V. The ac signal amplitude shall be less than 50 mV rms.

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3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.4.

3.2.4 Case outlines. The case outline shall be as specified in MIL-M-38510, appendix C, and 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in table III herein, for the applicable device type and device class. The electrical tests for each subgroup are described in table III, herein and table VI, test vectors which form a part of the manufacturing test type.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1 Total dose radiation hardness identifier. The total dose radiation hardness identifier shall be in accordance with MIL-M-38510, and as specified herein.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 105 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, as applicable, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D, using the circuit shown on figure 3, or equivalent.

2. T_A = +125°C.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) is specified as 10% for class B devices based on failures from group A, subgroups 1 and 7 tests, after cool down as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre-burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for the lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Constant acceleration in accordance with test condition D of method 2001 of MIL-STD-883.

e. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits specified in table II, footnote 3 herein, or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

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TABLE II. Electrical test requirements.

Line no.	MIL-STD-883 test requirements	Subgroups (see table III) Class B devices
1	Interim electrical parameters (method 5004)	1, 7
2	Dynamic burn-in (method 1015)	Required
3	Same as line 1	*1Δ 2/ 3/
4	Final electrical test parameters (method 5004)	*1, 2, 3, *7, 8, 9, 10, 11 1/ 4/
5	Group A test requirements (method 5005)	1, 2, 3, 4, 7, 8, 9, 10, 11, 5/
6	Group B end-point electrical parameters	See 4.4.2
7	Group C end-point electrical parameters (method 5005) 2/	1, 2, 3, 7, 8
8	Group D end-point electrical parameters (method 5005)	1, 2, 3, 7, 8

1/ (*) PDA applies to subgroups 1 and 7 (see 4.2c).

2/ (Δ) indicates a delta limit shall be required only on table III, subgroup 1 and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).

3/ Delta limit at 25°C for I_{CC} is 12 mA.

4/ The device manufacturer may, at his option, either complete the required electrical parameter measurements (subgroup 1), including delta measurements within 96 hours after burn-in completion (removal of bias); or may complete the required electrical measurements (subgroup 1) without delta measurements within 24 hours after burn-in completion (removal of bias).

5/ Capacitance testing (see 4.4.1b).

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be ≥ 2.0 V, L ≤ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1 DBE	2 RNEW/ SYNC	3 RD	4 WR	5 DOTW	6 ADS	7 RSTT	8 RSTO	9 RDY	10 PH12	11 PH11	12 GND	13 XIN
T_C = 25°C	I_{IL}	3009	1 2 3 4 5 6 7 8 9 10 11		0.0 V			0.0 V	0.0 V	0.0 V					0.0 V	0.0 V
	I_{IH}	3010	12 13 14 15 16 17 18 19 20 21 22		5.25 V			5.25 V	5.25 V	5.25 V						5.25 V
	I_{OZL} 1/		23 24				0.4 V	0.4 V								
	I_{OZH} 1/		25 26				5.25 V	5.25 V								
	V_{OL} 1/	3007	27 28 29 30 31 32 33 34 35 36 37		2 mA	2 mA	2 mA				2 mA	2 mA	1 mA	1 mA		
	V_{OH} 1/	3006	38 39 40 41 42 43 44 45 46 47 48		-1 mA	-1 mA	-1 mA				-1 mA	-1 mA	-1 mA	-1 mA		
	V_{T+} 1/ 2/	3006	49													
	V_{HTS} 1/ 3/	3006	50													
	I_{CC}	3005	51													4/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Test conditions (pins not designated may be ≥ 2.0 V, ≤ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	14 XOUT	15 FCLK	16 CTTL	17 TSO	18 WATT0	19 WATT1	20 WATT2	21 WATT3	22 CWATT	23 PER	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{IL}	3009	1											5.25 V	XIN	-10		mA
			2											"	RSTI	"		"
			3											"	RNEW/SYNC	"		"
			4											"	ADS	"		"
			5											"	DBIN	"		"
			6										0.0 V	"	PER	"		"
			7									0.0 V		"	CWAIT	"		"
			8								0.0 V			"	WAIT1	"		"
			9							0.0 V				"	WAIT2	"		"
			10						0.0 V					"	WAIT4	"		"
			11					0.0 V						"	WAIT8	"		"
	I_{IN}	3010	12											"	XIN		10	"
			13											"	RSTI		"	"
			14											"	RNEW/SYNC		"	"
			15											"	ADS		"	"
			16											"	DBIN		"	"
			17										5.25 V	"	PER		"	"
			18									5.25 V		"	CWAIT		"	"
			19								5.25 V			"	WAIT1		"	"
			20							5.25 V				"	WAIT2		"	"
			21						5.25 V					"	WAIT4		"	"
			22					5.25 V						"	WAIT8		"	"
	I_{OZL} 1/		23											"	RD	-20		"
			24											"	WR	-20		"
	I_{OZH} 1/		25											"	RD		20	"
			26											"	WR		20	"
	V_{OL} 1/	3007	27											"	RSTO		0.50	V
			28											"	RD		"	"
			29											"	WR		"	"
			30											"	DBE		"	"
			31				2 mA							"	TSO		"	"
			32											"	HOLT		"	"
			33		2 mA									"	FCLK		"	"
			34											"	PHI1		"	"
			35											"	PHI2		"	"
			36			2 mA								"	CTTL		"	"
			37	2 mA										"	XOUT		"	"
	V_{OH} 1/	3006	38											4.75 V	RSTO	4.275		V
			39											"	RD	"		"
			40											"	WR	"		"
			41											"	DBE	"		"
			42				-1 mA							"	TSO	"		"
			43											"	HOLT	"		"
			44		-1 mA									"	FCLK	"		"
			45											"	PHI1	"		"
			46											"	PHI2	"		"
			47			-1 mA								"	CTTL	"		"
			48	-1 mA										"	XOUT	"		"
	V_{T+} 1/ 2/	3006	49											5.00 V	RSTI	2.0	3.5	V
	V_{HYS} 1/ 3/	3006	50											5.00 V	RSTI	0.8	1.8	V
	I_{CC}	3005	51											5.25 V	V_{CC}		120	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designed may be H ≥ 2.0 V, L ≤ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1 DBE	2 WWEN/ SYNC	3 RD	4 WR	5 DOTN	6 AD5	7 RSYI	8 RSTO	9 RDY	10 PHI2	11 PHI1	12 GND	13 XIN
2 T_C = 125°C			52-102	Tests 52-102: Repeat tests 1 through 51, T_C = +125°C. For test limits, see table I.												
3 T_C = -55°C			103-153	Tests 103-153: Repeat tests 1 through 51, T_C = -55°C. For test limits, see table I.												
4 T_C = 25°C	C_O	3012	154 155 156 157 158 159 160 161 162 163 164	5/ 0.0 V	0.0 V	0.0 V 5/ 0.0 V	0.0 V 5/ 0.0 V	0.0 V	0.0 V	0.0 V	0.0 V 5/ 0.0 V	0.0 V 5/ 0.0 V	0.0 V 5/ 0.0 V	0.0 V 5/ 0.0 V	0.0 V	0.0 V
	C_I	3012	165 166 167 168 169 170 171 172 173 174 175	0.0 V	5/ 0.0 V			5/ 0.0 V	5/ 0.0 V	5/ 0.0 V			0.0 V	5/ 0.0 V		5/ 0.0 V
7 T_C = 25°C	6/ 7/ 8/	3014	176	Functional testing.												
8 T_C = 125°C T_C = -55°C	6/ 7/ 8/	3014	177	Functional testing. Same tests, terminal conditions and limits as subgroup 7 except T_C = 125°C and -55°C.												
9 T_C = 25°C	6/ 7/ 8/ 9/	3014	178-230	Switching tests (see table IV).												
10 T_C = 125°C	6/ 8/ 9/	3014	231-283	Switching tests, Repeat tests, 178 through 230 except T_C = 125°C												
11 T_C = -55°C	6/ 7/ 9/	3014	284-336	Switching tests, Repeat tests, 178 through 230 except T_C = -55°C												

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designed may be H $\geq$ 2.0 V, L $\leq$ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	14 XOUT	15 FCLK	16 CTTL	17 TSO	18 WATT8	19 WATT4	20 WATT2	21 WATT1	22 CWATT	23 PER	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2 T_C = 125°C			52-102	Tests 52-102: Repeat tests 1 through 51, T_C = +125°C														
3 T_C = -55°C			103-153	Tests 103-153: Repeat tests 1 through 51, T_C = -55°C														
4 T_C = 25°C	C_O	3012	154	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	DBE		40	pF
			155	"	"	"	"	"	"	"	"	"	"	"	RD		"	"
			156	"	"	"	"	"	"	"	"	"	"	"	WR		"	"
			157	"	"	"	"	"	"	"	"	"	"	"	RSTO		"	"
			158	"	"	"	"	"	"	"	"	"	"	"	RDY		"	"
			159	"	"	"	"	"	"	"	"	"	"	"	PHI2		"	"
			160	"	"	"	"	"	"	"	"	"	"	"	PHI1		"	"
			161	5/	"	"	"	"	"	"	"	"	"	"	XOUT		"	"
			162	0.0 V	5/	"	"	"	"	"	"	"	"	"	FCLK		"	"
			163	"	0.0 V	5/	"	"	"	"	"	"	"	"	CTTL		"	"
			164	"	"	0.0 V	5/	"	"	"	"	"	"	"	TSO		"	"
	C_I	3012	165	"	"	"	0.0 V	"	"	"	"	"	"	"	RWEN/SYNC		30	"
			166	"	"	"	"	"	"	"	"	"	"	"	DOUT		"	"
			167	"	"	"	"	"	"	"	"	"	"	"	ADS		"	"
			168	"	"	"	"	"	"	"	"	"	"	"	RSTT		"	"
			169	"	"	"	"	"	"	"	"	"	"	"	XIN		"	"
			170	"	"	"	"	"	"	"	5/	"	"	"	WATT1		"	"
			171	"	"	"	"	"	"	5/	0.0 V	"	"	"	WATT2		"	"
			172	"	"	"	"	"	5/	0.0 V	"	"	"	"	WATT4		"	"
			173	"	"	"	"	5/	0.0 V	"	"	"	"	"	WATT8		"	"
			174	"	"	"	"	0.0 V	"	"	"	5/	"	"	CWATT		"	"
			175	"	"	"	"	0.0 V	"	"	"	0.0 V	5/	"	PER		"	"
7 T_C = 25°C	6/ 7/ 8/	3014	176	Functional testing														
8 T_C = 125°C T_C = -55°C	6/ 7/ 8/	3014	177	Functional testing. Same tests, terminal conditions and limits as subgroup 7 except T_C = 125°C and -55°C														
9 T_C = 25°C	6/ 7/ 8/ 9/	3014	178-230	Switching tests (see table IV).														
10 T_C = 125°C	6/ 8/ 9/	3014	231-283	Switching tests, Repeat tests, 178 through 230 except T_C = 125°C														
11 T_C = -55°C	6/ 7/ 9/	3014	284-336	Switching tests, Repeat tests, 178 through 230 except T_C = -55°C														

See footnotes at end of table.

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1/ $V_{T(+)}$, V_{HYS}, V_{XH}, V_{OL}, I_{OZL}, I_{OZH} and V_{OH} are tested with the same conditions as subgroups 7 and 8.

2/ Increment input in 50 mV steps beginning 100 mV above the maximum limit specified until the output changes from GND to V_{CC}. The input voltage where this transition occurs is V_{T+}.

3/ $V_{HYS} = (V_{T+})-(V_{T-})$. See table I for V_{HYS} limits.

4/ Apply 20 MHz signal to XIN.

5/ The capacitance measurements should be made between the indicated terminal and ground at frequency of 1 MHz. The dc bias of the measuring instrument should be less than ±0.1 V. The ac signal amplitude shall be less than 50 mV rms. The tests will be conducted in accordance with MIL-STD-883, method 3012.

6/ For functional tests, input/output conditions A, B, and C, appear in table V.

7/ Table VI functional tests shall be repeated for input/output conditions A, B, and C of table V (see 6.3).

8/ When patterns are applied as specified for subgroups 7 and 8, they verify the dynamic, functional output threshold and switching parameters at T_C = +25°C, -55°C, and +125°C (see 6.3).

9/ All subgroups 9, 10, and 11 tests are run concurrently with subgroups 7 and 8.

10/ $\overline{RSTO}$ goes high on the first rising edge of PHI1 after $\overline{RSTI}$ goes high.

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TABLE IV. Switching tests for group A inspection for device type 01.

Test number	Symbol	Terminals	Figure 5	Reference number 1/	Min	Max	Units
178	t_{Cp}	PHI1	A	9	100	48	ns
179	t_{CLh}	PHI1	A	10	35	47	ns
180	t_{CLl}	PHI1	A	13	43	60	ns
181	$t_{CLw(1,2)}$	PHI1/PHI2	A	54 & 53	40	52	ns
182	t_{CLwas}	PHI1/PHI2			-5	5	ns
183	t_{CLR}	PHI1	A	12		7	ns
184	t_{CLF}	PHI1	A	11		7	ns
185	t_{nOVL}	PHI2	A	14	-2	5	ns
186	t_{nOVLas}	PHI1/PHI2			-4	4	ns
187	t_{Xh}	XIN	A	1	16		ns
188	t_{Xl}	XIN	A	2	16		ns
189	t_{XFr}	XIN/FCLK	A	3	6	29	ns
190	t_{XFf}	XIN/FCLK	A	4	6	29	ns
191	t_{XCr}	XIN/CTTL	A	5	6	34	ns
192	t_{XPr}	XIN/PHI1	A	8	3	32	ns
193	t_{FCr}	FCLK/CTTL	A	6	2	6	ns
194	t_{FCf}	FCLK/CTTL	A	7	-3	4	ns
195	t_{FPr}	FCLK/PHI1	B	16	-7	4	ns
196	t_{FPf}	FCLK/PHI1	B	17	-4	-3	ns
197	$t_{CWh(W)}$	PHI2/$\overline{CWAIT}$	H	44	20	30	ns
198	t_{PCf}	PHI2/CTTL	B	18	-3	4	ns
199	t_{CTw}	CTTL	A	20	43	51	ns
200	t_{PCr}	PHI1/CTTL	B	19	-3	5	ns
201	t_{CTr}	CTTL	B	21		5	ns
202	t_{CTF}	CTTL	B	22		5	ns
203	t_{RSTr}	PHI1/$\overline{RSTO}$	F	31		21	ns
204	t_{RSTs}	$\overline{RSTI}$/PHI1	F	30	20		ns
205	t_{ADs}	$\overline{ADS}$/PHI1	F	32	30		ns
206	t_{ADw}	$\overline{ADS}$	F	33	25		ns
207	t_{DDs}	$\overline{DDIN}$/PHI1	F	34	15		ns
208	t_{Tf}	PHI1/$\overline{TSO}$	C & E	23		12	ns
209	t_{Tr}	PHI1/$\overline{TSO}$	C & E	24		18	ns
210	$t_{RWF(F)}$	PHI1/$\overline{RD}$,$\overline{WR}$	E	25		35	ns
211	$t_{RWf(S)}$	PHI1/$\overline{RD}$,$\overline{WR}$	C	35		25	ns
212	t_{RWr}	PHI1/$\overline{RD}$,$\overline{WR}$	C & E	26		25	ns
213	$t_{DBf(W)}$	PHI1/$\overline{DBE}$	C & E	27		28	ns
214	$t_{DBf(R)}$	PHI2/$\overline{DBE}$	C & E	28		21	ns
215	t_{DBr}	PHI2/$\overline{DBE}$	C & E	29		23	ns
216	t_{PLz}	(RWEN/SYNC)/($\overline{WR}$ & $\overline{RD}$)	D	36		30	ns
217	t_{PHz}	(RWEN/SYNC)/($\overline{WR}$ & $\overline{RD}$)	D	39		30	ns
218	t_{PZL}	(RWEN/SYNC)/($\overline{WR}$ & $\overline{RD}$)	D	37		25	ns
219	t_{PZH}	(RWEN/SYNC)/($\overline{WR}$ & $\overline{RD}$)	D	38		25	ns
220	$t_{CWs(H)}$	$\overline{CWAIT}$/PHI1	G	41	30		ns
221	$t_{CWh(H)}$	PHI1/$\overline{CWAIT}$	G	42	0		ns
222	$t_{CWs(W)}$	$\overline{CWAIT}$/PHI2	G & H	40	10		ns

See footnote at end of table.

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TABLE IV. Switching tests for group A inspection for device type 01 - Continued.

Test number	Symbol	Terminals	Figure 5	Reference number 1/	Min	Max	Units
223	t_{Ws}	$\overline{WAITn}$/PHI2	H	45	7		ns
224	t_{Wh}	PHI2/$\overline{WAITn}$	H	46	20		ns
225	t_{Ps}	$\overline{PER}$/PHI1	I	49	7		ns
226	t_{Ph}	PHI1/$\overline{PER}$	I	48	30		ns
227	t_{Ph}	PHI2/RDY	H & I	47		25	ns
228	t_{Sgs}	($\overline{RWEN}$/SYNC)/FCLK	J	50	6		ns
229	t_{Syh}	FCLKN/$\overline{RWEN}$/SYNC	J	51	0		ns
230	t_{Cs}	CTTL/($\overline{RWEN}$/SYNC)	J	52		10	ns

1/ The reference number refers to the position where the parameter being measured appears on figure 5.

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TABLE V. Input and output conditions for table III for device type 01.

Symbol	Figure 5	Reference number 1/	Terminals	Test conditions 2/			Unit
				A	B	C	
V_{CC}			V_{CC}	5.25	4.75	5.25	V
V_{OH}			All logic outputs except XOUT	2.0	2.0	2.0	V
V_{OL}			All logic outputs except XOUT	1.6	1.6	1.6	V
V_{IH}			All logic inputs except $\overline{RSTI}$ and XIN	3.0	3.0	5.75	V
V_{IL}			All logic inputs except $\overline{RSTI}$ and XIN	0.4	0.4	-0.5	V
V_{T+}				4.2	4.2	4.2	V
V_{HYS}				1.80	1.80	1.80	V
V_{XH}			XIN	4.7	4.2	5.75	V
V_{XL}			XIN	0.5	0.5	-0.5	V
t_{CP}	A	9	PHI1	100	100	100	ns
t_{CLh}	A	10	PHI1	35	35	35	ns
t_{CLl}	A	13	PHI1	60	60	60	ns
$t_{CLw(1,2)}$	A	53 & 54	PHI1 and PHI2	52	52	52	ns
t_{CLR}	A	12	PHI1	8	8	8	ns
t_{CLF}	A	11	PHI1	8	8	8	ns
t_{nOVL}	A	14	PHI1 and PHI2	-2	-2	-2	ns

See footnotes at end of table.

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TABLE V. Input and output conditions for table III for device type 01 - Continued.

Symbol	Figure 5	Reference number 1/	Terminals	Test conditions 2/			Unit
				A	B	C	
t_{Xh}	A	1	XIN	16	16	16	ns
t_{X1}	A	2	XIN	16	16	16	ns
t_{XFr}	A	3	XIN/FCLK	29	29	29	ns
t_{XFf}	A	4	XIN/FCLK	29	29	29	ns
t_{XCr}	A	5	XIN/CTTL	34	34	34	ns
t_{XPr}	A	8	XIN/PHI1	32	32	32	ns
t_{FCr}	A	6	FCLK/CTTL	6	6	6	ns
t_{FCf}	A	7	FCLK/CTTL	4	4	4	ns
t_{FPr}	B	16	PHI1/FCLK	4	4	4	ns
t_{FPf}	B	17	FCLK/PHI1	2	2	2	ns
t_{PCf}	B	18	PHI2/CTTL	4	4	4	ns
t_{PCr}	B	19	PHI1/CTTL	5	5	5	ns
t_{CTR}	B	21	CTTL	5	5	5	ns
t_{CTF}	B	22	CTTL	5	5	5	ns
t_{RSTr}	F	31	PHI1/$\overline{RSTO}$	21	21	21	ns
t_{RSTs}	F	30	PHI1/$\overline{RSTI}$	20	20	20	ns

See footnotes at end of table.

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TABLE V. Input and output conditions for table III for device type 01 - Continued.

Symbol	Figure 5	Reference number 1/	Terminals	Test conditions 2/			Unit
				A	B	C	
t_{ADs}	F	32	PHI1/$\overline{ADS}$	30	30	30	ns
t_{ADw}	F	33	$\overline{ADS}$	25	25	25	ns
t_{DDs}	F	34	PHI1/$\overline{DDIN}$ (T2)	15	15	15	ns
t_{Tf}	C & E	23	PHI1/$\overline{TSO}$	12	12	12	ns
t_{Tr}	C & E	24	PHI1/$\overline{TSO}$	18	18	18	ns
$t_{RWf(F)}$	E	25	PHI1/($\overline{RD}$/$\overline{WR}$)	35	35	35	ns
$t_{RWf(S)}$	C	35	PHI1/($\overline{RD}$/$\overline{WR}$)	25	25	25	ns
t_{RWr}	C & E	26	PHI1/($\overline{WR}$/$\overline{RD}$)	25	25	25	ns
$t_{DBf(W)}$	C & E	27	PHI1/$\overline{DBE}$	28	28	28	ns
$t_{DBf(R)}$	C & E	28	PHI2/$\overline{DBE}$	21	21	21	ns
t_{DBr}	C & E	29	PHI2/$\overline{DBE}$	23	23	23	ns
t_{PLZ}	D	36	($\overline{RWEN}$/SYNC)/($\overline{WR}$ & $\overline{RD}$)	30	30	30	ns
t_{PHZ}	D	39	($\overline{RWEN}$/SYNC)/($\overline{WR}$ & $\overline{RD}$)	30	30	30	ns
t_{PZL}	D	37	($\overline{RWEN}$/SYNC)/($\overline{WR}$ & $\overline{RD}$)	25	25	25	ns
t_{PZH}	D	38	($\overline{RWEN}$/SYNC)/($\overline{WR}$ & $\overline{RD}$)	25	25	25	ns
$t_{CWs(H)}$	G	41	$\overline{CWAIT}$/PHI1	30	30	30	ns
t_{CTV}	B	20	CTTL	43	43	43	ns

See footnotes at end of table.

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TABLE V. Input and output conditions for table III for device type 01 - Continued.

Symbol	Figure 5	Reference number 1/	Terminals	Test conditions 2/			Unit
				A	B	C	
$t_{CWh(H)}$	G	42	PHI1/$\overline{CWAIT}$	0	0	0	ns
$t_{CWs(W)}$	G & H	40	$\overline{CWAIT}$/PHI2	10	10	10	ns
$t_{CWh(W)}$	H	44	PHI2/$\overline{CWAIT}$	20	20	20	ns
t_{Ws}	H	45	$\overline{WAITn}$/PHI2	7	7	7	ns
t_{Wh}	H	46	PHI2/$\overline{WAITn}$	20	20	20	ns
t_{Ps}	I	49	$\overline{PER}$/PHI1	7	7	7	ns
t_{Ph}	I	48	PHI1/$\overline{PER}$	30	30	30	ns
t_{Rd}	H & I	47	PHI2/RDY	25	25	25	ns
t_{SYs}	J	50	($\overline{RWEN}$/SYNC)/FCLK	6	6	6	ns
t_{SYh}	J	51	FCLK/($\overline{RWEN}$/SYNC)	0	0	0	ns
t_{CS}	J	52	CTTL/($\overline{RWEN}$/SYNC)	10	10	10	ns

1/ The reference numbers refer to the position where the parameter being measured appears on figure 5.

2/ Test conditions:

A - - - V_{CC} = 5.25 V, f = 10 MHz

B - - - V_{CC} = 4.75 V, f = 10 MHz

C - - - V_{CC} = 5.25 V, f = 10 MHz
V_{IH} = 5.75 V, and V_{IL} = -0.5 V

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Device type 01

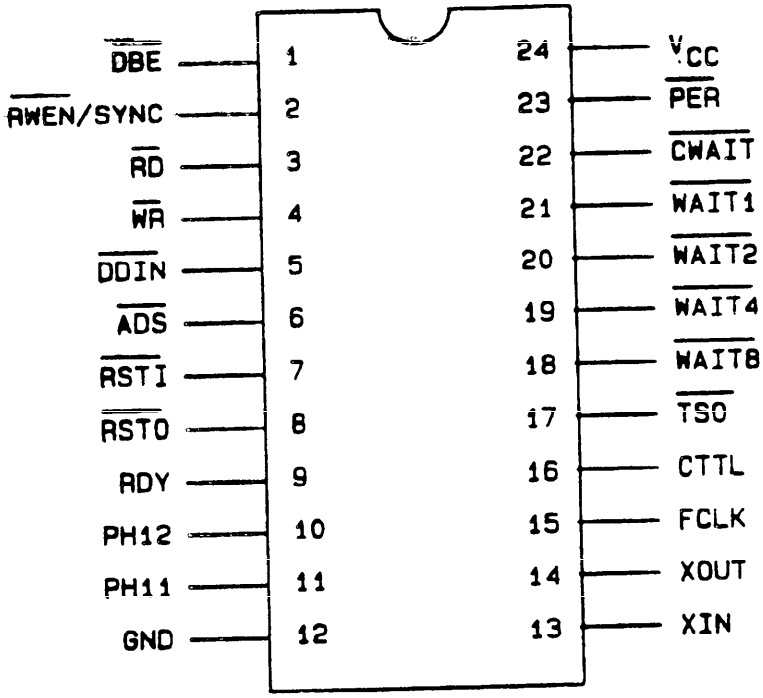


FIGURE 1. Terminal connections.

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Device type 01

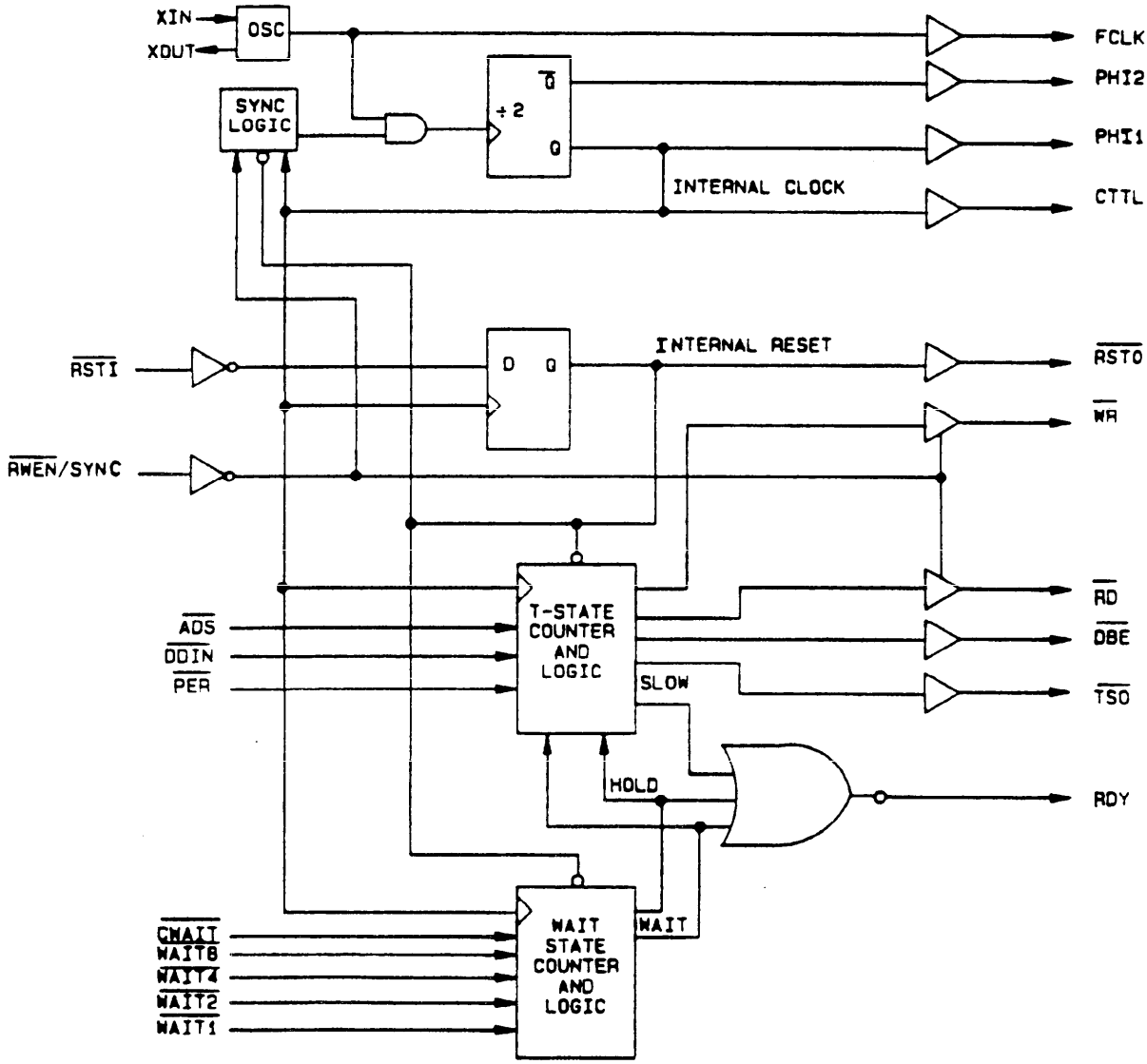
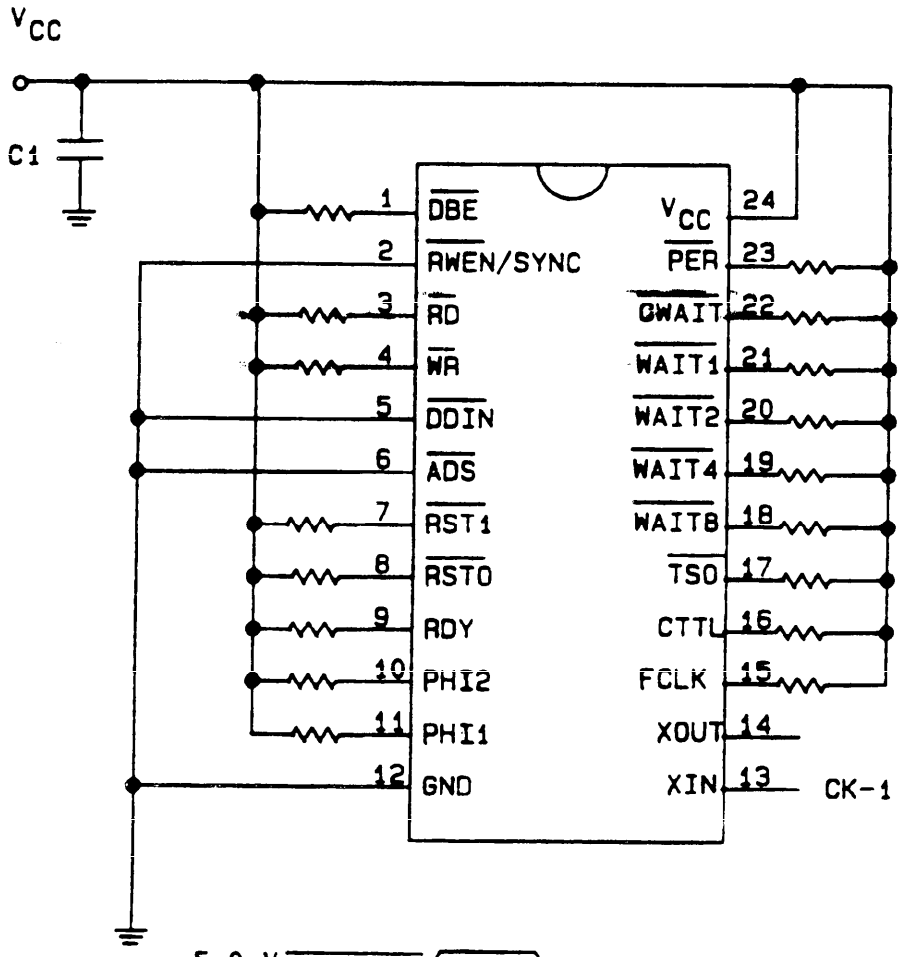
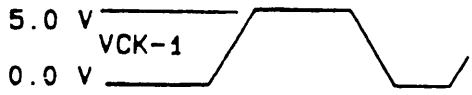


FIGURE 2. Functional block diagram.

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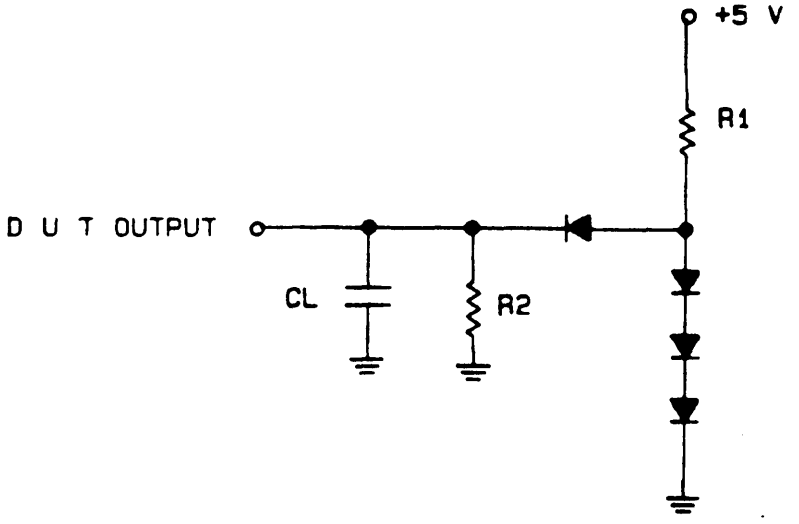


CK-1 = 1.0 MHz ± 10%
50% duty cycle.

Notes:
1. C1 = 0.1 µF ±10%.
2. R1 = 2.7 kΩ ±10%.
3. I_{CC} = 120 mA.
4. V_{CC} = 5.25 V.
5. Power/Socket = 0.65 W.
6. $V_{CK-1} \leq V_{CC}$.
7. Both V_{CC} and V_{CK-1} are sequenced. (V_{CC} should be turned on before V_{CK-1} and to be turned off in reversed order.)

FIGURE 3. Dynamic burn-in and life test circuit.

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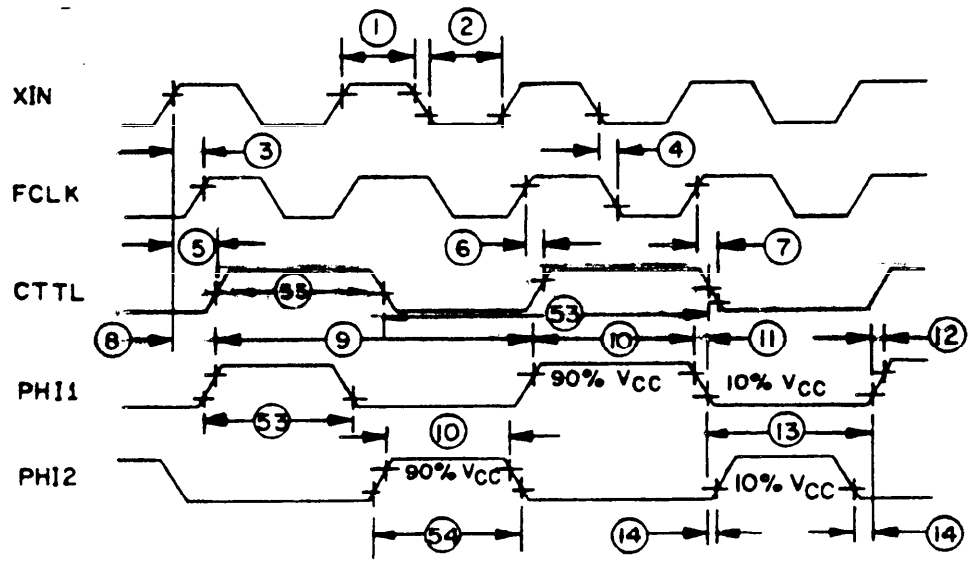


R1 = 1.90 kΩ
R2 = 4.5 kΩ
CL = 50 pF on RDY, $\overline{DBE}$, $\overline{TSO}$, CTTL
CL = 75 pF on $\overline{WR}$, $\overline{RD}$
CL = 100 pF on FCLK
CL = 170 pF on PHI1, PHI2
CL = includes all stray capacitance

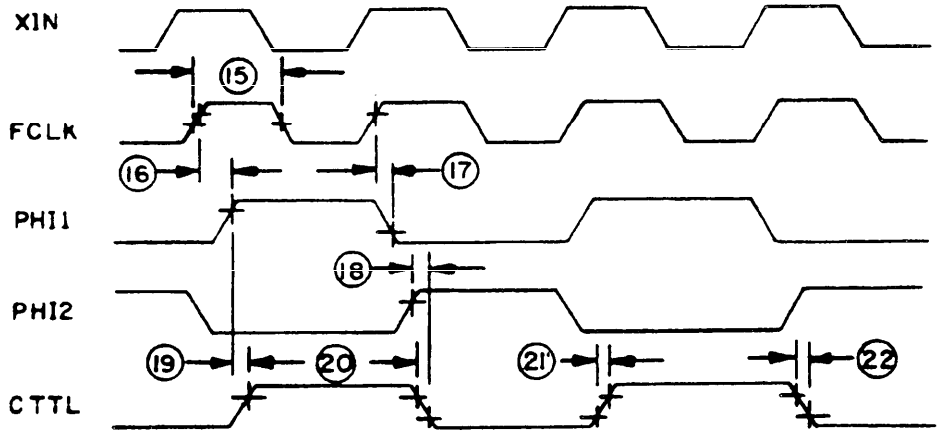
NOTE: Equivalent circuit may be used.

FIGURE 4. Output load circuit for ac, functional, and three-state tests.

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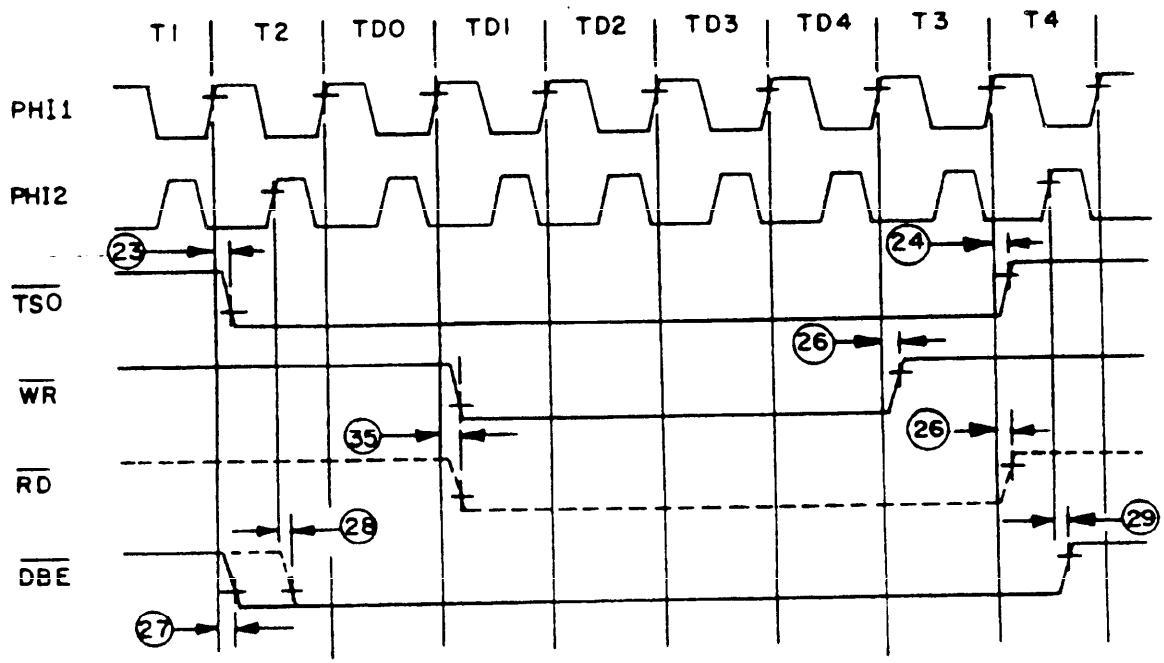
(A) Clock signals



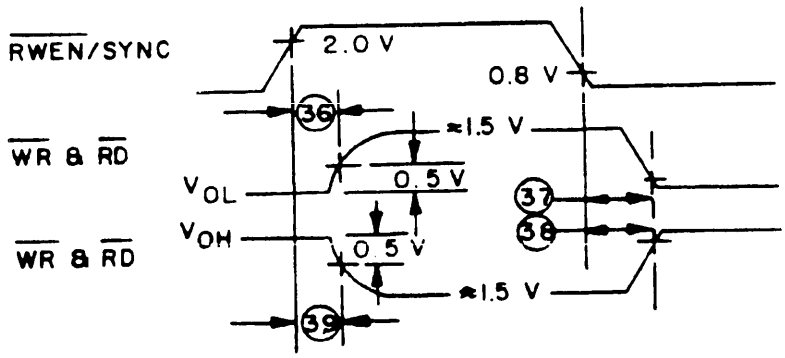
(B) Clock signals

FIGURE 5. Timing diagrams.

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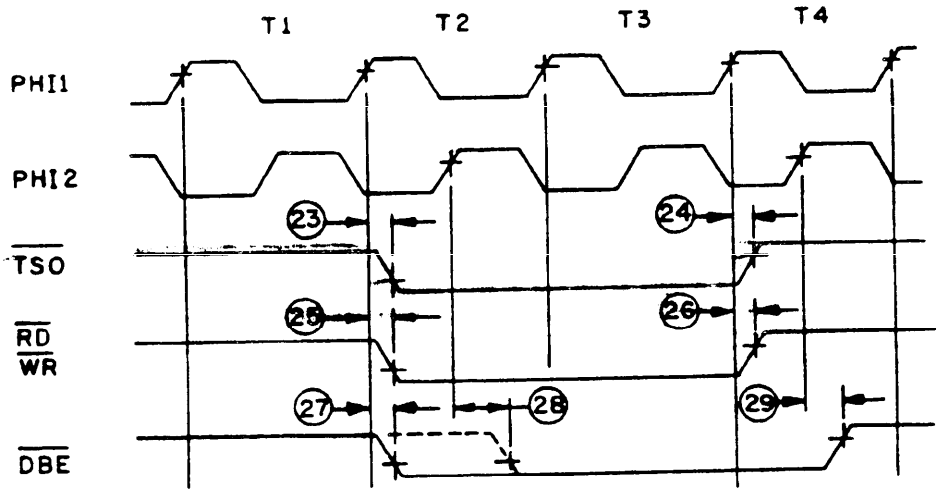
(C) Control outputs (peripheral cycle)



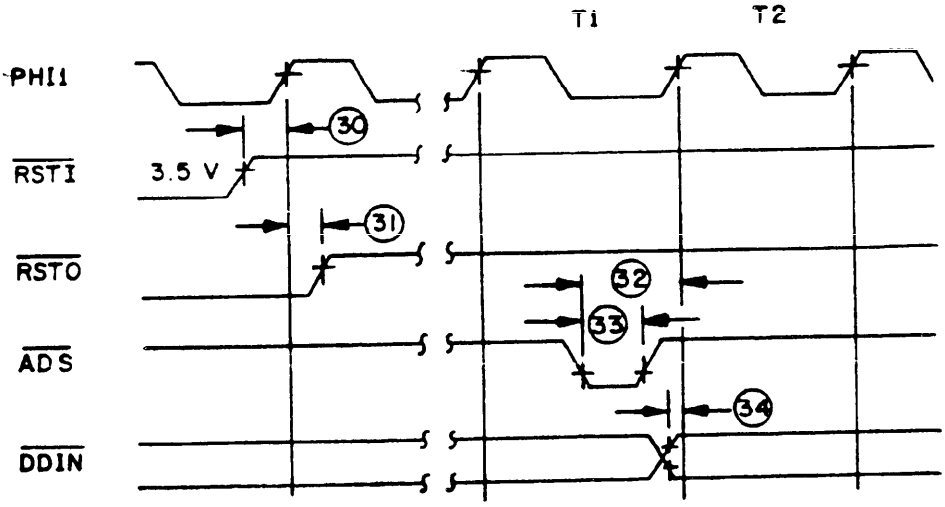
(D) Control outputs (three-state timing)

FIGURE 5. Timing diagrams - Continued.

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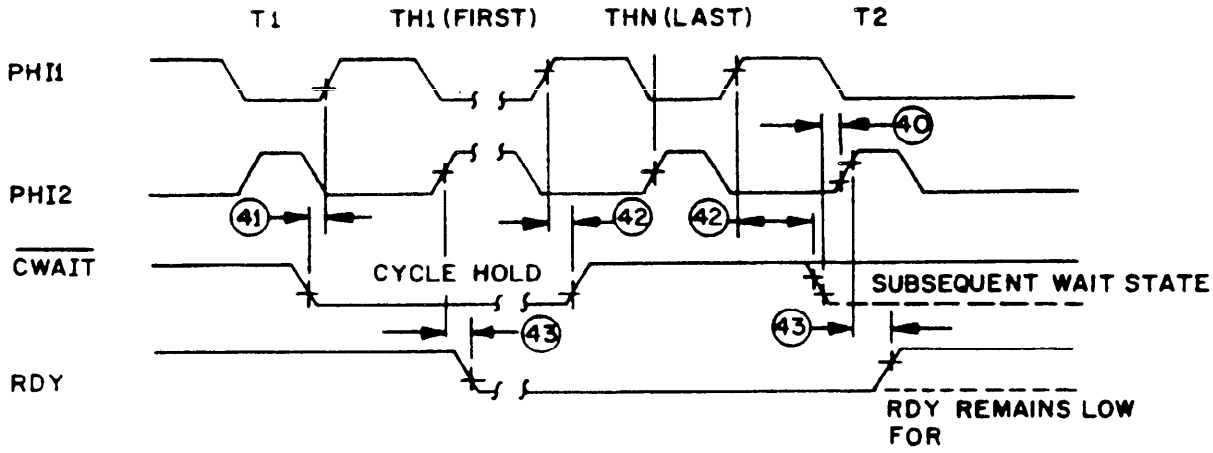
(E) Control inputs (fast cycle)



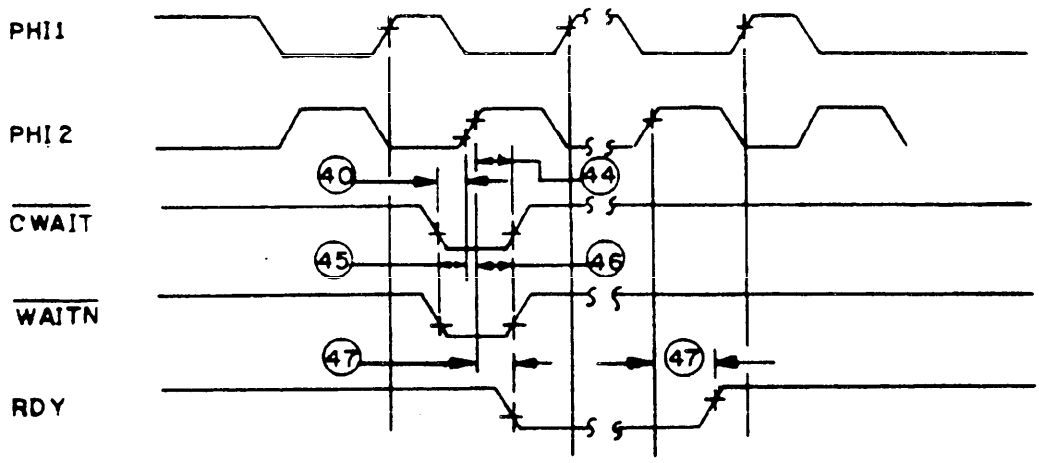
(F) Control inputs

FIGURE 5. Timing diagrams - Continued.

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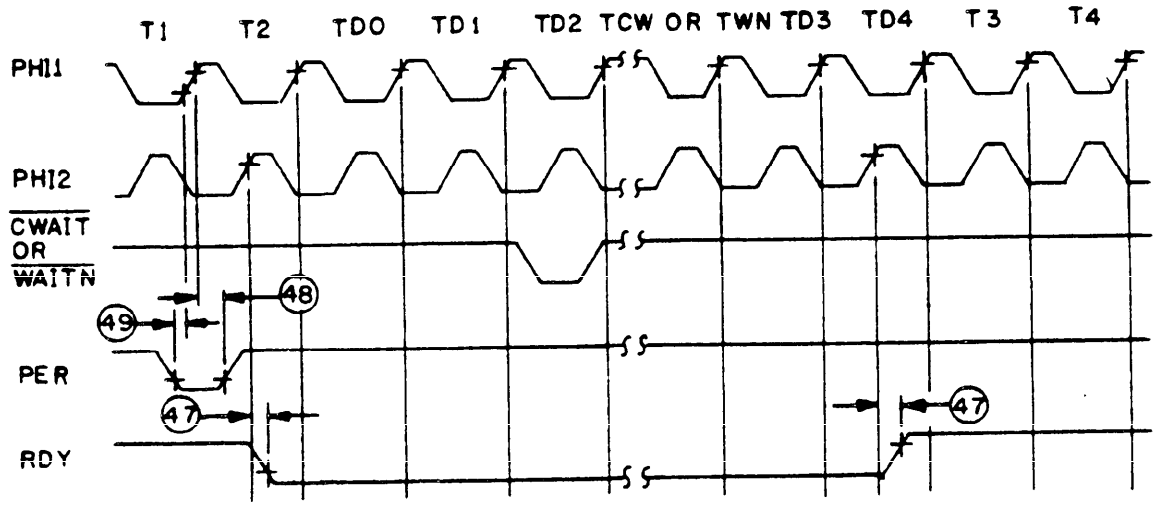
(G) Cycle hold



(H) Wait state (fast cycle)

FIGURE 5. Timing diagrams - Continued.

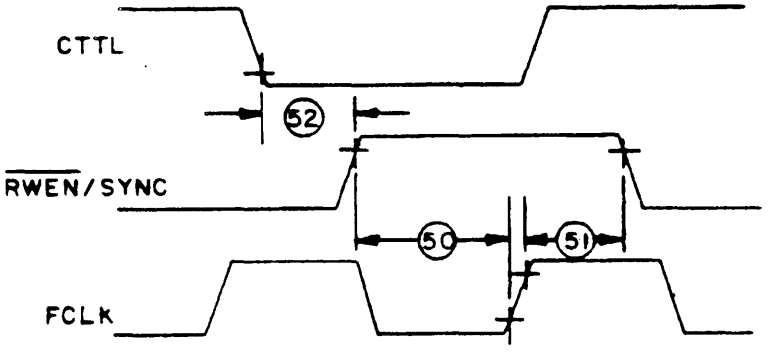
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(I) Wait state (peripheral cycle)

FIGURE 5. Timing diagrams - Continued.

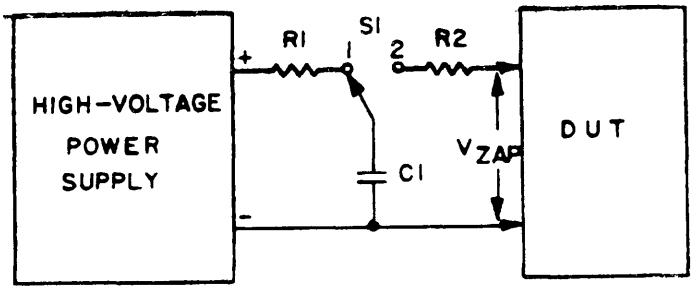
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(J) Synchronization timing

FIGURE 5. Timing diagrams - Continued.

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NOTES:

1. R1 = 3 Gohm maximum to 800 kΩ minimum.
2. R2 = 1.5 kΩ ±10%.
3. C1 = 100 pF ±10%.
4. S1 = Hg-wetted 'bounceless' relay break before make type.
5. All pins of DUT no associated with V_{ZAP} test shall be open (NC).
6. Devices shall be tested to MIL-STD-883, method 3015 requirements (see 4.6.2).

FIGURE 6. High voltage (V_{ZAP}) test circuit.

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TABLE VI. Test vectors.

Table VI, not printed herein, is a computer listing for use with table III. Each vector identifies activity states by pin number. See 6.3 for information on obtaining the complete table in a compatible output format. When the vectors are performed as specified in table III for subgroups 7, 8, 9, 10, and 11 they verify the dynamic, functional, threshold, and switching parameters at +25°C, -55°C, and +125°C.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.4). In addition, the qualification outlined below shall be performed through the use of the identified qualification vehicle.

Qualification phase	Qualification vehicle
Control and stability	Process monitor die (PM)

4.3.1 Process control and stability. Process control and stability of dc parameters must be demonstrated through the use of the manufacturer's process monitor (PM). The PM (either a drop-in or fabricated in the kerf) is to be designed so that the dc process parameters (ac parameters may be included as a manufacturer's option) may be measured in wafer form or packaged device form. The PM design must be submitted to the qualifying activity for approval prior to use for qualification and must contain as a minimum the following structures:

N channel device (minimum geometry)
P channel device (minimum geometry)
Sheet resistance measurement structure
Metal step coverage structure
Field threshold device
Intermetal oxide integrity structure
Contact chains (to be of sufficient length to be representative of the contact resistance typically found on a device, with diagnostic taps to isolate failures).
  Metal to poly
  Metal to diffusion

For qualification, PM's on a minimum of three different lots (minimum of four PM's per wafer) shall be measured to ensure the establishment of a statistically valid data base upon which a decision can be made as to whether the manufacturer's process is stable and under control.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.4).

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4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and quality/accept number values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroup 4(C_D, C_I and C_{CLK}) capacitance measurements shall be performed only for initial qualification and after process or design changes which may affect device capacitance.

c. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

d. Dynamic and functional testing shall be performed using circuit configuration shown on figure 4, utilizing the waveforms on figure 5.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table IIB of method 5005 of MIL-STD-883, and as follows: A special subgroup 9 shall be added to the group B inspection requirements for class B. This subgroup shall consist of the test, conditions, and limits specified in 4.6.2.

4.4.3 Groups C and D inspection. Groups C and D inspections shall consist of the test subgroups and LTPD values shown in tables III and IV of method 5005 of MIL-STD-883 and as follows.

a. End-point electrical tests shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883).

1. Test condition D using the circuit shown on figure 3.

2. T_A = +125°C minimum.

3. Test duration, 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

c. Constant acceleration in accordance with test condition D of method 2001 of MIL-STD-883.

4.4.4 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883.

4.5 Inspection of packaging. Inspection of packaging shall be as specified in MIL-M-38510.

4.6 Methods of inspection. Methods of inspection shall be as follows.

4.6.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and are positive when flowing into the referenced terminal.

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4.6.2 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 V from a 100 pF source in the following test sequence:

a. V_{ZAP} test circuit is shown on figure 6.

b. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

c. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 2-pulse sequence as follows and in accordance with the test circuit shown on figure 6.

(1) Input (-) to GND.

(2) Input (+) to GND.

d. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. In addition, all devices shall be in contact with a conductive material which shorts all leads together to prevent electrostatic damage (see 6.6).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Acquisition requirements. The acquisition document must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete Part or Identifying Number (see 6.7).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to qualifying activity, if applicable.

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g. Requirement for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

h. Requirements for product assurance options.

i. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the PIN. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

k. Requirements for "JAN" marking.

6.3 Test vector table. Table VI (test vectors) which forms a part of this specification is not printed herein because of its extreme length and complexity. When the test vectors are presented in hardcopy form, it requires conversion to online format. When the entire sequence of vector groups are needed for an inspection, a test program output should be obtained in a form compatible with test system architecture. The preparing activity, or Defense Electronics Supply Center as its agent, may be consulted for obtaining such output information. For reference purposes, Rome Air Development Center (RADC-RBRM) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited quantity of dated, hardcopy printouts of this table.

6.4 Functional description, terms and symbols. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

6.4.1 Functional description. The timing control unit (TCU) is a 24-pin device using a CMOS technology. It provides a two-phase clock, system control logic, and cycle extension for compatible 32-bit microprocessors. The TCU input clock can be provided by either a crystal or an external clock signal whose frequency is twice the system clock frequency.

The two-phase clock (PHI1 and PHI2) are used mainly for the central processing unit (CPU) and memory management unit (MMU); also, the TCU produces two system clocks (FCLK and CTTL) for general use within the system. FCLK is a fast clock whose frequency is the same as the input clock (20 MHz), while CTTL is a replica of PHI1. The system control logic ($\overline{RSTO}$, $\overline{WR}$, $\overline{RD}$, $\overline{DBE}$, $\overline{TSO}$, and RDY) provides accurate bus control signals and very controllable bus cycle timing.

6.4.2 Terms and definitions.

$\overline{RSTI}$ (Reset, input)	Active low. Schmit triggered asynchronous signal used to generate system reset.
$\overline{ADS}$ (Address strobe)	Active low. Identifies the first timing state (T1) of a bus cycle.
$\overline{DDIN}$ (Data direction)	Active low. Indicates the direction of the data transfer during a bus cycle. Implies a read when low and a write when high.
$\overline{RWEN}$/SYNC (Read/Write enable and synchronization)	3-state the $\overline{RD}$ and $\overline{WR}$ outputs when high and enables them when low. Also used to synchronize the phase of the TCU clock signals, when two or more TCU's are used.
XIN (Crystal or external clock source)	Input from a crystal or an external clock source.

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$\overline{\text{CWAIT}}$ (Continuous wait)	Active low. Initiates a continuous wait if sampled when low in the middle of T2. If $\overline{\text{CWAIT}}$ is low at the end of T1, it initiates a cycle hold.
$\overline{\text{WAIT1}}$, $\overline{\text{WAIT2}}$, $\overline{\text{WAIT4}}$, $\overline{\text{WAIT8}}$ (4-bit wait state)	Active low. These inputs (collectively called $\overline{\text{WAITn}}$) allow from zero to 15 wait states to be specified. They are binary weighted.
$\overline{\text{PER}}$ (Peripheral cycle)	Active low. If active, causes the TCU to insert five wait cycles into a normal bus cycle. It also causes the read and write signals to be reshaped to meet the setup and hold timing requirements of slower MOS peripherals.
$\overline{\text{RSTO}}$ (Reset output)	Active low. This signals becomes active when $\overline{\text{RSTI}}$ is low, initiates a system reset. $\overline{\text{RSTO}}$ goes high on the first rising edge of PHI1 after $\overline{\text{RSTI}}$ goes high.
$\overline{\text{RD}}$ (Read strobe)	3-state active low. Identifies a read cycle. It is decoded from $\overline{\text{DDIN}}$ and 3-state by $\overline{\text{RWEN}}$/SYNC.
$\overline{\text{WR}}$ (Write strobe)	3-state active low. Identifies a write cycle. It is decoded from $\overline{\text{DDIN}}$ and TRI-STATE by $\overline{\text{RWEN}}$/SYNC.
$\overline{\text{DBE}}$ (Data bus enable)	Active low. This signal is used to control data bus buffers. It is low when the data buffers are to be enabled.
$\overline{\text{TSO}}$ (Timing state output)	Active low. The falling edge of $\overline{\text{TSO}}$ signals the beginning of T2 of a bus cycle. The rising edge of $\overline{\text{TSO}}$ signals the beginning of T4.
RDY (Ready)	Active high. This signal will go low and remain low as long as wait states are to be inserted in a bus cycle.
FCLK (Fast clock)	This is a clock running at the same frequency as the crystal or external source. Its frequency is twice that of the CPU clocks.
PHI1 and PHI2 (CPU clocks)	These outputs provide the CPU with two-phase, non-overlapping clock signals. Their frequency is half that of the crystal or the external source.
CTTL (TTL system clock)	This is a system version of the PHI1 clock. Hence it operates at the CPU clock frequency.
XOUT (Crystal output)	This line is used as a return path for the crystal (if used). It must be open when an external clock source is used to drive XIN.

6.5 Logistic support. Lead material and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for government logistic support will be acquired to device class B for the Department of Defense (see 1.2.2), and lead finish "C" (see 3.3). Longer length leads and lead forming shall not affect the PIN.

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6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. These CMOS devices are fabricated with silicon gate technology including input protection, which reduces the susceptibility to damage; however, the following practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Devices should be stored in conductive foam or carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50%, if practical.

6.7 Part or Identifying Number (PIN). The PIN shall be in accordance with MIL-M-38510.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry part. Generic-industry microcircuit types may not have equivalent operational performance characteristics, due to military temperature ranges, reliability factors equivalent to MIL-M-38510 device type, or slight differences in case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	32C201-10

CONCLUDING MATERIAL

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Navy - EC
Air Force - 17

Review activities:
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Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1136)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

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c. Reason/Rationale for Recommendation:

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